Fatemeh Naeinian

New York University, Department of Electrical and Computer Engineering
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Education

New York University **Jan 2025 – present**
PhD in Electrical Engineering *New York City, USA*
- GPA : 3.9/4

University of Tehran **Sep 2019 – Jun 2024**
B.Sc in Electrical Engineering *Tehran, Iran*

University of Tehran **Sep 2021 – Jun 2024**
Minor in Computer Engineering *Tehran, Iran*

Publications

- **F. Naeinian**, A. Hamza, H. Zhu, A. Choromanska, "Zero-Shot Cross-City Generalization in End-to-End Autonomous Driving: Self-Supervised versus Supervised Representations", arXiv preprint arXiv:2603.11417.
- **F. Naeinian**, E. Balazadeh, M. Tale Masouleh, "Mapping Human Grasping to 3-Finger Grippers: A Deep Learning Perspective", *32nd International Conference on Electrical Engineering (ICEE)*, pp. 1–7, 2024.

Research Interests

- End-to-end autonomous driving systems
- Cross-domain generalization
- Self-supervised and representation learning
- Motion forecasting and trajectory prediction
- Multi-modal perception and sensor fusion
- World Models

Research Experience

Graduate Research Assistant, Learning Systems Laboratory **New York University**
Advisor: Prof. Anna Choromanska *Jan 2025 – present*
- **Zero-Shot Cross-City Generalization in End-to-End Autonomous Driving: Self-Supervised versus Supervised Representations** [arXiv]
  Investigated zero-shot geographic generalization in end-to-end autonomous driving by evaluating self-supervised and supervised visual representations under strict city-based splits on nuScenes and NAVSIM.
- **TALKtoME: Educational Materials for Speech and Language Acquisition in Autism** [Project Website]
  Contributed to the development of a curated visual dataset of images and videos representing action verbs and verb–noun pairs, designed to support speech and language acquisition in autistic children and adults.

B.Sc Thesis **University of Tehran**
TaarLab: The Human and Robot Interaction Lab *Jul. 2023 – Mar. 2024*
- Built a human hand-feature dataset using MediaPipe and trained object detection models to map grasping behavior to optimal grasping points for a 3-finger gripper.

Internship **Remote**
Technical University of Munich *Summer 2022*
- Conducted Boundary Computation for determining the Reachable Set of Robotic Manipulators.

Poster Presentation

- End-to-End Autonomous Driving, Computer Vision Day, New York, April 2026.

Leadership & Mentorship

- **Mentor, NYU Arise Program**, New York, NY Summer 2025
  Supervised a team of high school students on a machine learning project, guiding them through dataset preparation, model training, and result evaluation. Mentored students in scientific communication, leading to a successful poster presentation at the program's closing event.

Academic Service

- Reviewer, IEEE Robotics and Automation Letters (RA-L), 2026.

Relevant Coursework

New York University

- Advanced Machine Learning
- Applied Matrix Theory
- Digital Signal Processing
- Deep learning
- Computer Vision
- Probability and Stochastic Process

University of Tehran

- Neural Networks
- Machine Learning
- Artificial Intelligence
- Data Structures and Algorithms
- Fundamentals of Mechatronics Engineering
- Advanced programming

Honors and Awards

- Received the School of Engineering (SoE) PhD Fellowship at NYU for an academic year, 2025.
- Ranked within the Top 20% among more than 116 B.Sc. students of the Electrical Engineering Department at the University of Tehran, 2023.
- Awarded the Supporter Foundation of the University of Tehran Honorable Award for Academic Excellence, 2019-2021.
- Ranked 3rd in the Control branch of Electrical Engineering, 2023.
- Ranked among the top 0.1% (92nd) in the nationwide university entrance exam in Mathematics and Physics fields for a B.Sc. degree, 2019.
- Ranked among the top 0.1% (91th) in the nationwide university entrance exam in Foreign Languages fields for a B.Sc. degree, 2019.
- Gold Prize in Tebyan Student Projects Festival in computer seminars, 2017.

Skills

Programming Languages & Frameworks :

- Proficient in Python, MATLAB, C, C++, Verilog
- Familiar with TensorFlow, NumPy, Pandas, Pyomo, Matplotlib, Keras, scikit-learn, PyTorch, Seaborn, CSS, HTML, JavaScript

Softwares & Developer Tools :

- Familiar with Jupyter Notebook, Google Colab, ModelSim, Quartus II, MultiSim, STM32CubeIDE, LaTeX, Visual Studio Code, Arduino

Teaching Experience

Course Assistant at New York University:

- Digital Signal Processing (Fall 2025)

Teaching Assistant at the University of Tehran:

- Neural Networks and Deep Learning (Spring & Fall 2023)
- Linear Control Systems (Fall 2023)
- Probability and Statistics (Spring 2023)
- Linear Control Systems (Fall 2022)
- Electrical Machines (Fall 2021)
- Engineering Mathematics (Fall 2022 & Spring & Fall 2023)
- Operational Research (Fall 2023)
- Linear Algebra (Fall 2022)
- Electronics I (Spring 2021)
- Introduction to Electrical Engineering (Spring 2021)

Languages

- **English** [Advanced Proficiency]
- **Persian** [Native]